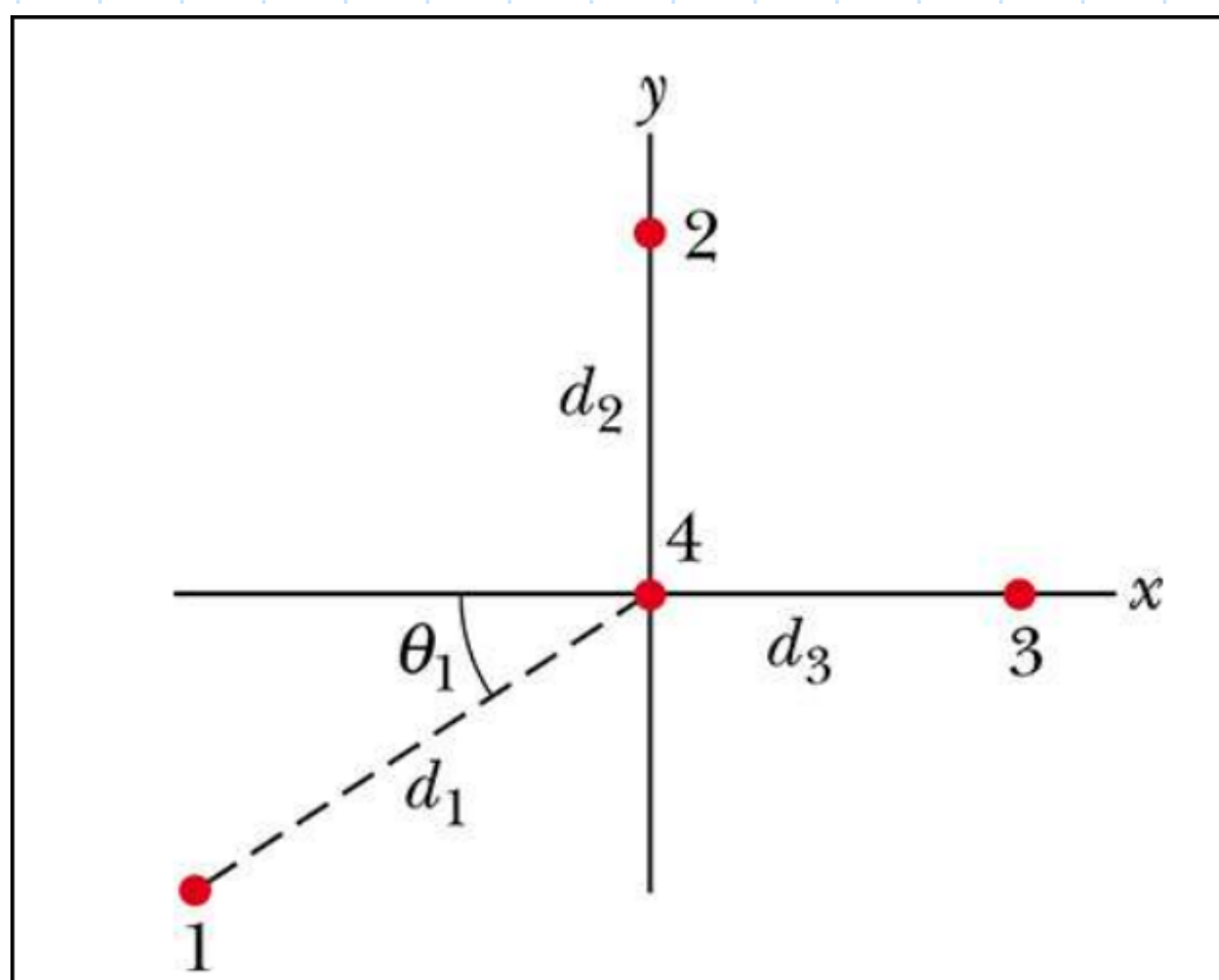


Pr-1



In Fig., what are the (a) magnitude and (b) direction of the net electrostatic force on particle 4 due to the other three particles? All four particles are fixed in the xy plane, and $q_1 = -3.20 \times 10^{-19} \text{ C}$, $q_2 = +3.20 \times 10^{-19} \text{ C}$, $q_3 = +6.40 \times 10^{-19} \text{ C}$, $q_4 = +3.20 \times 10^{-19} \text{ C}$, $\theta_1 = 35.0^\circ$, $d_1 = 3.00 \text{ cm}$, and $d_2 = d_3 = 2.00 \text{ cm}$.

Pr-2

*15. Deimos is a small moon of Mars, with a mass of $2 \times 10^{15} \text{ kg}$. Suppose that an electron is at a distance of 100 km from Deimos. What is the gravitational attraction acting on the electron? What negative electric charge would have to be placed on Deimos to balance this gravitational attraction? How many electron charges does this amount to? Treat the mass and the charge distributions as pointlike in your calculations.

Pr-3

33. A small, 2.00-g plastic ball is suspended by a 20.0-cm-long string in a uniform electric field as shown in Figure P23.33. If the ball is in equilibrium when the string makes a 15.0° angle with the vertical, what is the net charge on the ball?

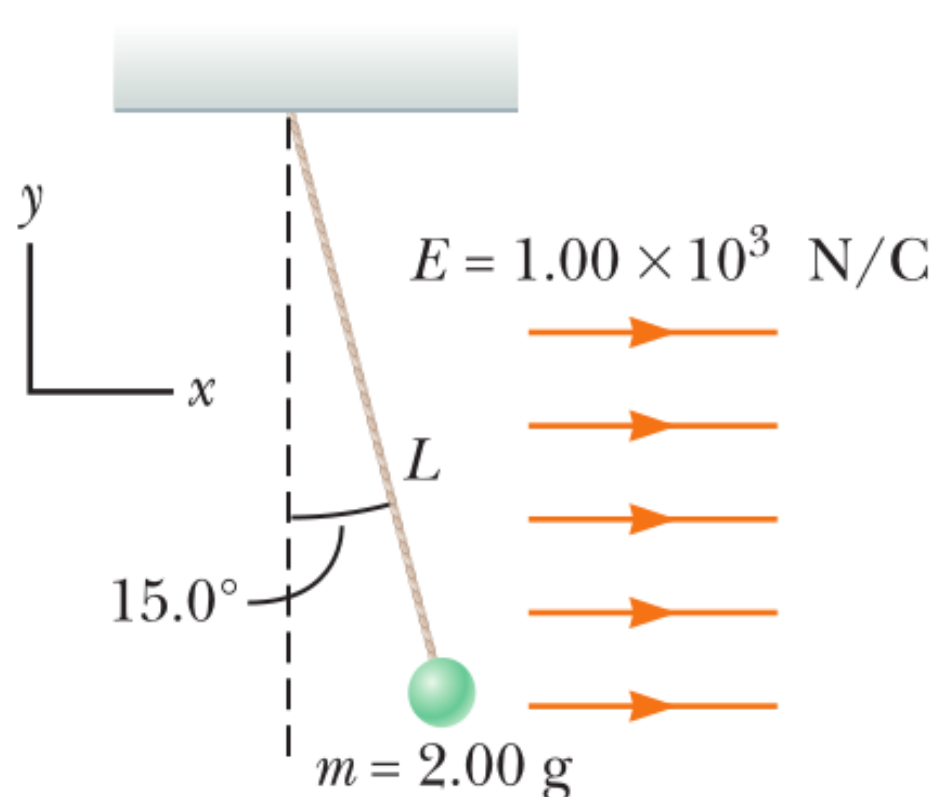


Figure P23.33